TRACKING & COUNTING

How the project handles difficult cases ? and where the limits are

OCCLUSION

Deep SORT keeps track memory for about six seconds, while OSNet appearance features help reconnect a person after a track break. Re-identification also checks movement distance and similarity. This is best-effort: a person hidden too long may return with a new ID or no usable crossing. In the example, the child disappeared behind a woman and his OUT was missed.

GROUPS ENTERING TOGETHER

There is no group-level shortcut: detections are tracked and counted per person. A crossing must be observed for an individual track and confirmed beyond the line. People entering close together can therefore count separately when visible, but overlap can hide a person, merge detections, or swap IDs and cause missed or incorrect events.

RE-ENTRY

Global identities and appearance prototypes are kept during one video run. When a new local track appears, the code tries to match it to an inactive identity using appearance similarity (at least 0.72), spatial distance, and an ambiguity margin. A matched person can then generate another IN/OUT event. This is not persistent across separate program runs and is not guaranteed after long absences.

STAFF VS. CUSTOMERS

The code attempts to identify two staff members from fixed startup regions during the first two seconds, requiring repeated detections. Their appearance profiles can recover staff tracks in a designated staff-side region after 12 consistent matches. Confirmed staff are excluded from customer counting; the starting customer baseline excludes the two identified staff. This camera-specific setup is not a general staff classifier.